

TIC 140918728

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.822098 d
Epoch (t0) 1517.7160 BTJD
Duration 0.0840 d
Depth 1.680 % (16,797 ppm)
SNR 230.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

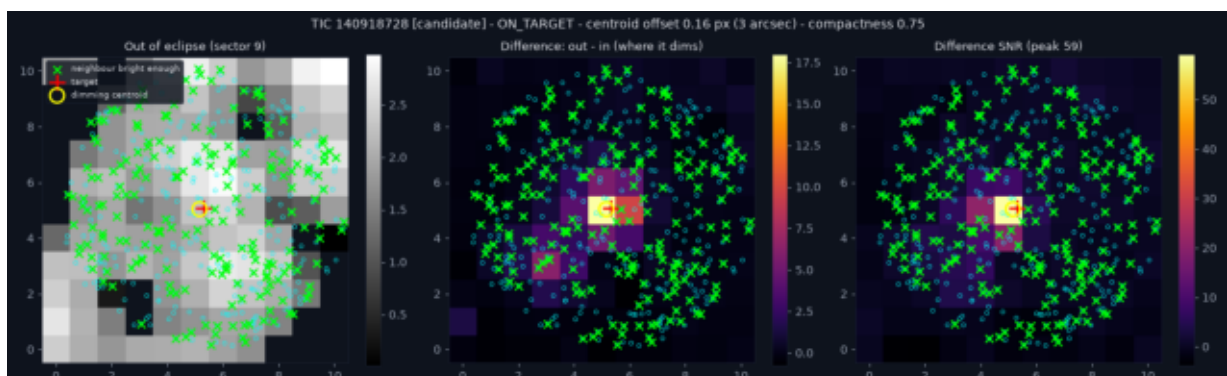
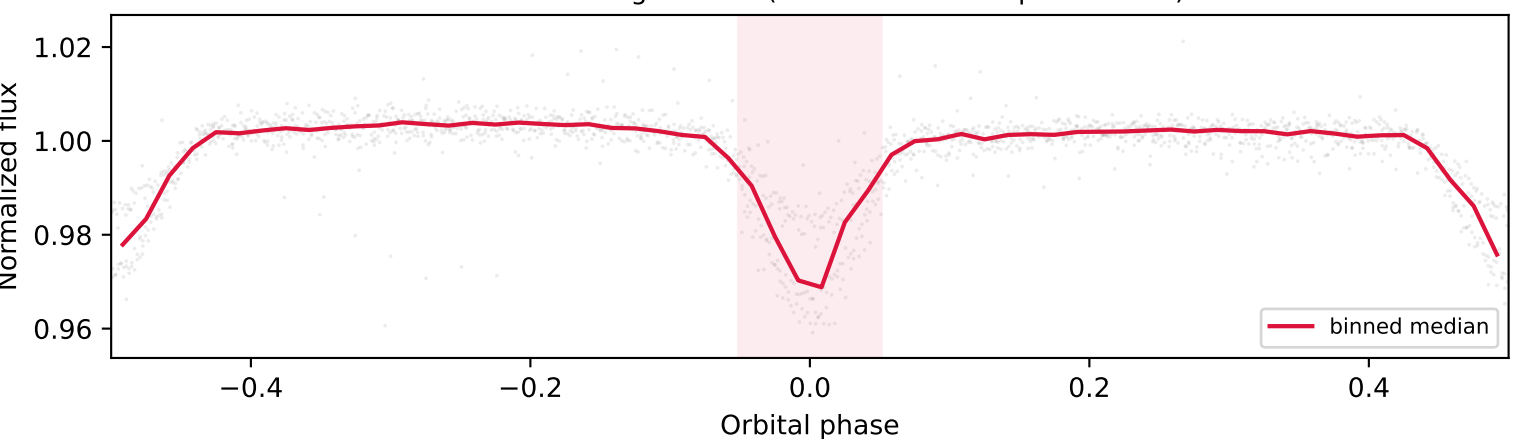
Eclipses observed 51 (gate ≥ 3)
Robust depth ratio 1.08 (gate ≥ 0.50)
Robust depth sigma 71.8 (gate ≥ 7.0)
Odd/even z-score 0.66 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 9 light-curve dip in this sector: 93.9 sigma t0 refined by 0.0050 phase
Difference-image peak SNR 59.15 (gate ≥ 5.0) Compactness 0.75 (gate ≥ 0.50)
Dimming centroid offset 0.16 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 819235642
0.53 px away, dTmag +3.38 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.